

Q1. A fair die is rolled. The probability of getting a number greater than 4 is:

a) $1/2$

b) $2/3$

c) $1/3$

d) $5/6$

Q2. If $P(A) = 0.7$ and $P(B) = 0.5$, and $P(A \cap B) = 0.4$, then $P(A \cup B)$ is:

a) 0.6

b) 0.7

c) 0.8

d) 0.9

Q3. If two events A and B are independent, then:

a) $P(A|B) = P(B)$

b) $P(A \cap B) = P(A) P(B)$

c) $P(A \cap B) = P(A) + P(B)$

d) $P(A \cup B) = P(A) P(B)$

Q4. The sum of probabilities of all mutually exclusive and exhaustive events is:

a) 0

b) 1

c) Less than 1

d) Greater than 1

Q5. Which of these is NOT a property of PMF?

a) $\sum p(x) = 1$

b) $p(x) \geq 0$ for all x

c) Defined for continuous variables

d) Represents discrete probability distribution

Q6. The expectation of a continuous random variable X is:

- a) $\sum x p(x)$
- b) $\int_{-\infty}^{\infty} xf(x)dx$
- c) $\sum p(x)$
- d) Undefined

Q7. If X is a discrete random variable, then $P(X = x)$:

- a) May be negative
- b) Is between 0 and 1
- c) Can be greater than 1
- d) Must be zero

Q8. If X has PMF $P(X=0) = 0.3$, $P(X=1) = 0.4$, $P(X=2) = 0.3$, $E[X]$ is:

- a) 1.0
- b) 1.5
- c) 2.0
- d) 0.9

Q9. If X is a discrete random variable with $E(X) = 5$, then:

- a) The most likely value is 5
- b) X must be equal to 5
- c) The average value of X over many trials is 5 ✓
- d) The probability that $X = 5$ is 1

Q10. Which function gives the probability that a random variable is less than or equal to a certain value?

- a) PDF
- b) PMF
- c) CDF
- d) MGF